

Assumptions, diagnostics, and multicollinearity

Last time: Variance inflation factors

Motivation: High multicollinearity means a predictor can be estimated well by some combination of other predictors

Variance inflation factor: Let X_i be a predictor in the model, and R_i^2 the coefficient of determination for a model estimating X_i from the other predictors. The *variance inflation factor* of X_i is

$$VIF_i = \frac{1}{1 - R_i^2}$$

- + Higher VIF means more multicollinearity
- + Rule of thumb: worried when $VIF_i > 5$ (i.e. $R_i^2 > 0.8$)

Class activity

+ $\widehat{\text{Hgt97}} = 40.59 + 1.10 \text{ Hgt96} \quad R^2_{adj} = 0.949$

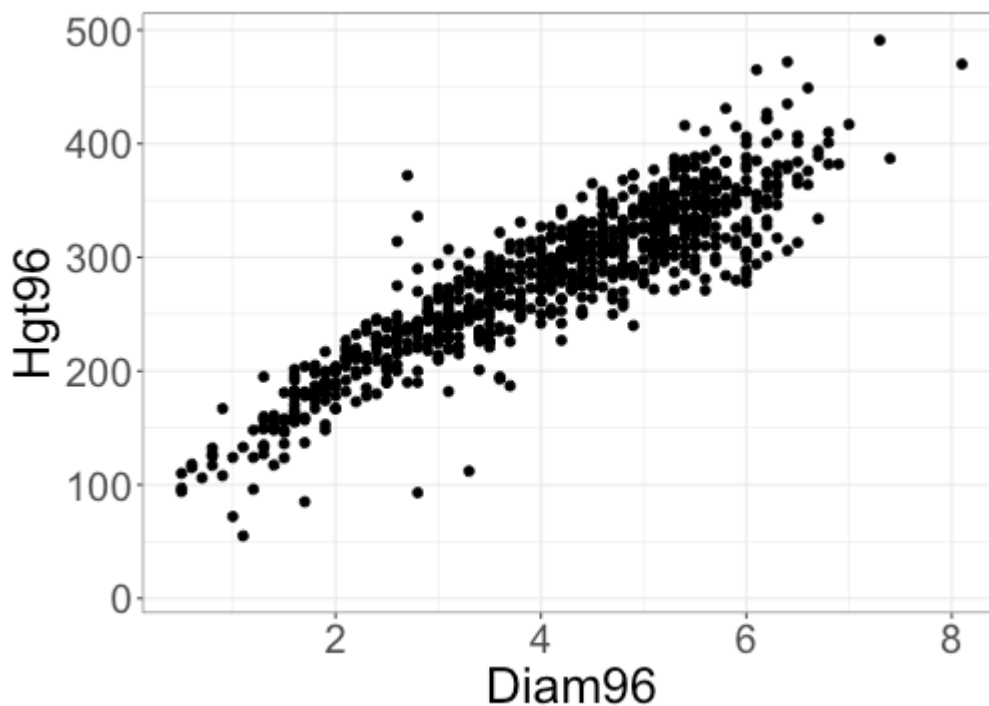
+

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96} \quad R^2_{adj} = 0.942$$

Why doesn't R^2_{adj} increase when I add diameter to the model?

Class activity

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Because diameter and height in 1996 are so correlated.

Class activity

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96}$$

What is the variance inflation factor for Diam96?

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What is the variance inflation factor for Diam96?

```
m1 <- lm(Diam96 ~ Hgt96, data = Pines)
summary(m1)$r.squared
```

```
## [1] 0.8134518
```

$$VIF = \frac{1}{1 - 0.813} = 5.35$$

Why is multicollinearity a problem?

Interpretation of individual terms is difficult!

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Does this mean that tree diameter is negatively associated with height?

Why is multicollinearity a problem?

Interpretation of individual terms is difficult!

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96}$$

Does this mean that tree diameter is negatively associated with height?

No! Model with just diameter:

$$\widehat{\text{Hgt97}} = 161.77 + 46.13 \text{ Diam96}$$

Problem: Diameter and Height in 1996 are strongly correlated; doesn't make sense to fix one and change the other.

Why is multicollinearity a problem?

It is hard to estimate the coefficients!

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It is hard to estimate the coefficients!

Model with just height in 1996:

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	40.590784	2.524484393	16.07884	5.283829e-51
##	Hgt96	1.096059	0.008734402	125.48762	0.000000e+00

Model with both height and diameter in 1996:

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	40.244053	3.04659425	13.209522	2.598076e-36
##	Hgt96	1.115861	0.02173417	51.341300	2.057358e-260
##	Diam96	-1.256481	1.02359067	-1.227523	2.199725e-01

Project data

- + MED_EARNINGS: median earnings for graduates from each university
- + CONTROL: whether the school is public or private
- + SATVRMID: midpoint of SAT critical reading scores of students attending the school
- + SATMTMID: midpoint of SAT math scores
- + SATWRMID: midpoint of SAT writing scores
- + ACTCMMID: midpoint of the ACT cumulative scores
- + ACTENMID: midpoint of ACT English scores
- + ACTMTMID: midpoint of ACT math scores
- + + others...

Which variables might exhibit multicollinearity?

Project data

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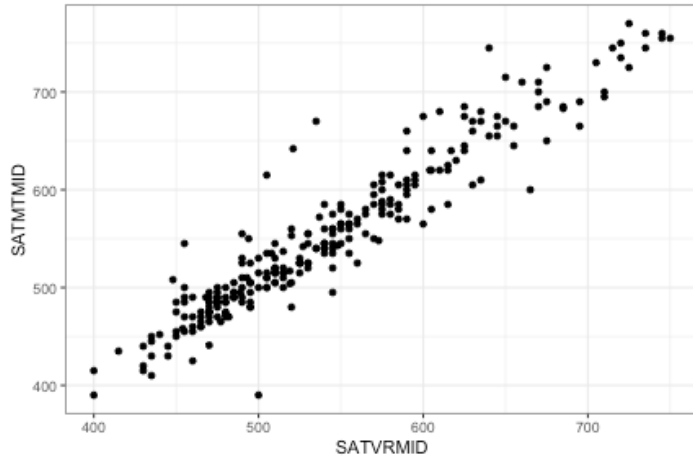
There is probably multicollinearity with all the test score variables

Project data

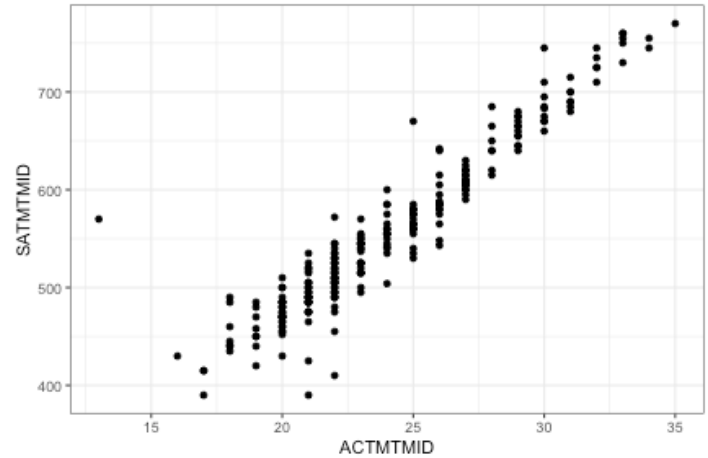
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How can we visually investigate correlated predictors?

Project data



$$r = 0.95$$



$$r = 0.94$$

VIFs in R

Fortunately, R makes it easy to calculate VIFs:

```
library(car)
m1 <- lm(MED_EARNINGS ~ SATVRMID + SATMTMID +
          SATWRMID + ACTCMMID + ACTENMID +
          ACTMTMID + ACTWRMID,
          data = scorecard)
vif(m1)
```

```
## SATVRMID SATMTMID SATWRMID ACTCMMID ACTENMID ACTMTMID AC
## 29.78240 17.69560 27.89062 27.68832 21.90292 17.26265 1
```

- + The `vif` function (from the `car` package) calculates a VIF for each predictor
- + The variance inflation factor for SAT verbal score is 29.78, e.g.

Handling multicollinearity

We have a few options for handling multicollinearity:

- + Remove variables
 - + Example: just pick one of the test score variables, and discard the rest, since they are so highly correlated

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- + Create a new variable
 - + Example: take an average or sum of the test score variables, instead of using all of them individually
- + Ignore it
 - + If we don't care about interpreting coefficients or doing inference, multicollinearity doesn't matter as much

Class activity

Work on the class activity (in R).

Class activity

```
rho <- 0
X <- MASS::mvrnorm(n, mu = c(0, 0), Sigma = matrix(c(1, rho, rho, 1), 2, 2))
x1 <- X[,1]
x2 <- X[,2]

y <- 1 + x1 + x2 + rnorm(n)
m1 <- lm(y ~ x1)
m2 <- lm(y ~ x1 + x2)
coef(m1)
```

```
## (Intercept)          x1
##      1.124596      1.073064
```

```
coef(m2)
```

```
## (Intercept)          x1          x2
##      1.1129517      1.0202536      0.9645505
```

Class activity

```
rho <- 0.9
X <- MASS::mvrnorm(n, mu = c(0, 0), Sigma = matrix(c(1, rho, rho, 1), 2, 2))
x1 <- X[,1]
x2 <- X[,2]

y <- 1 + x1 + x2 + rnorm(n)
m1 <- lm(y ~ x1)
m2 <- lm(y ~ x1 + x2)
coef(m1)
```

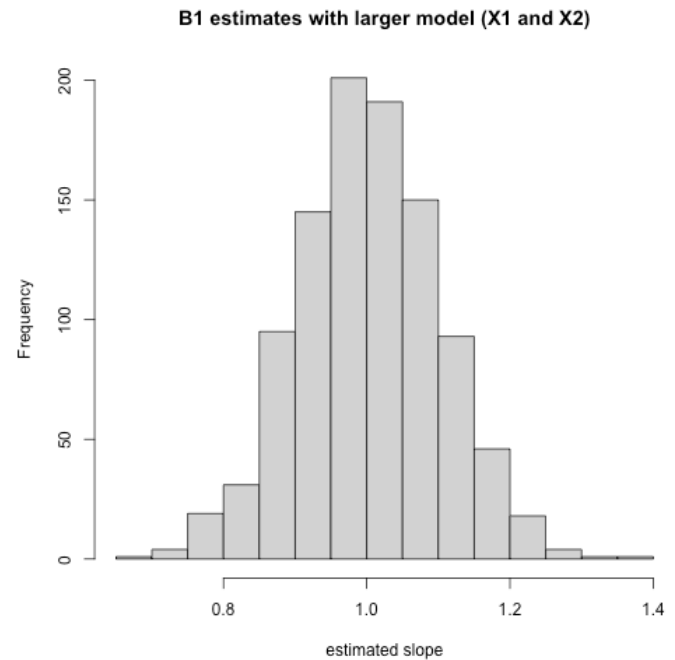
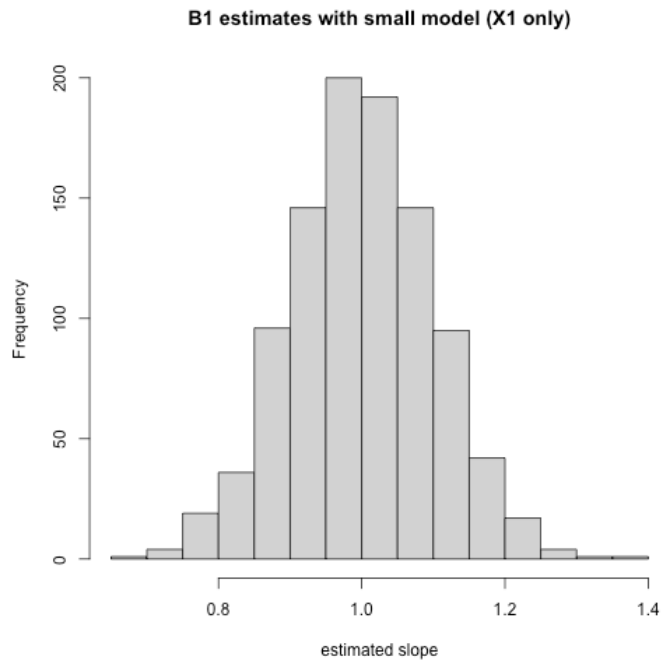
```
## (Intercept)          x1
##    0.9361277    1.8706363
```

```
coef(m2)
```

```
## (Intercept)          x1          x2
##    0.9446137    1.3926972    0.5789134
```

Class activity

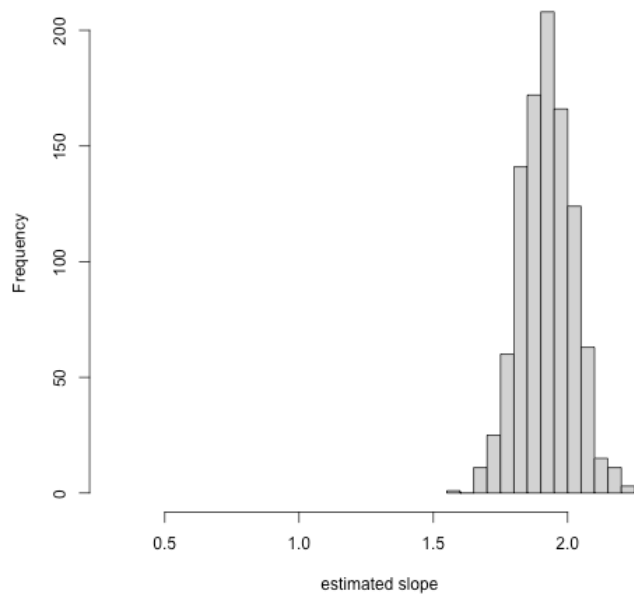
When correlation = 0:



Class activity

When correlation = 0.9:

B1 estimates with small model (X1 only)



B1 estimates with larger model (X1 and X2)

